Newborough Warren

What’s Happening to the Water Under the Dunes?

A 21-year groundwater study at one of Wales’s most important coastal dune sites

Hollingham, M. (2026). Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales. Draft, 2026.
Project: github.com/newbroman/Newborough_Hydrology
Web tools (scenario viewer, flood forecaster): newbroman.github.io/Newborough_Hydrology
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The place

Newborough Warren sits at the southern tip of Anglesey, covering roughly 1,300 hectares of sand dunes, wetlands and conifer plantation. It is designated as a Special Area of Conservation — one of the most important coastal dune systems in Wales.

The site is home to rare plant communities called dune slacks: low-lying hollows between the dunes where groundwater rises close to the surface each winter, creating seasonally flooded wetlands that support creeping willow, orchids and a rich variety of wetland invertebrates.

The survival of these habitats depends entirely on groundwater. If the water table sits too low in summer, the plant communities that define dune slacks shift irreversibly towards drier grassland. The difference between a thriving wet slack and a degraded dry one is remarkably small — just 37 centimetres of water table depth in summer.

A large portion of the northern dunes — around 700 hectares — was planted with Corsican pine between 1948 and 1965. This plantation intercepts roughly a quarter of incoming rainfall before it can reach the ground, reducing the amount of water that recharges the aquifer beneath.

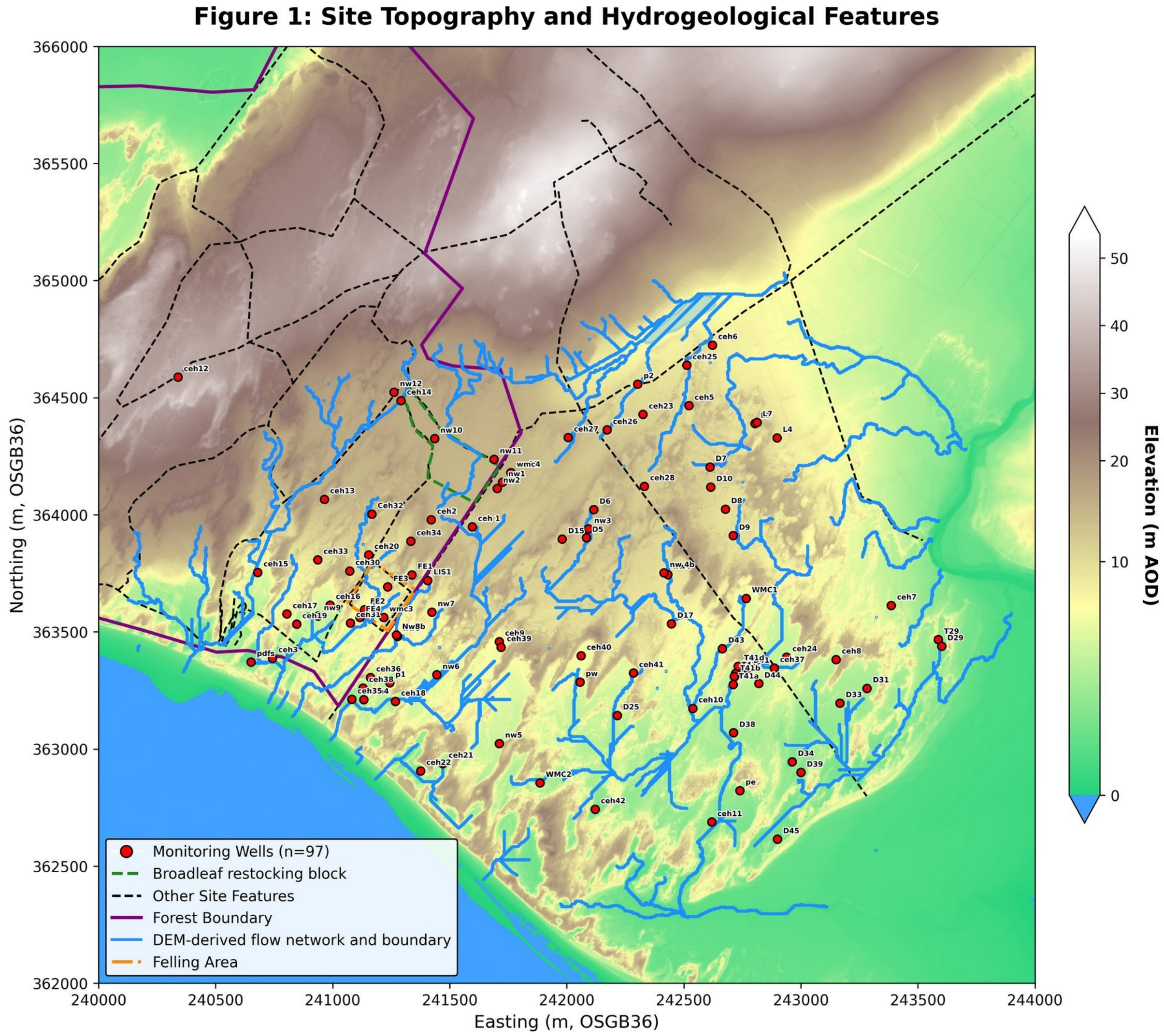

Figure 1. Site overview showing the topography of Newborough Warren, the monitoring network, the forest boundary (magenta), the 2017 felling area (orange), and the drainage network derived from the digital elevation model. The map plots all surveyed dipwell locations; 88 wells passed quality control and entered the analysis.

The study

The study monitored 88 wells across the site for 21 years (2005–2026), measuring water levels monthly with simple hand-held instruments called dipwells. Combined with publicly available rainfall and temperature data from a nearby weather station, this network was analysed using a 43-step data pipeline to understand how the groundwater system works, how it is changing, and what management can realistically achieve.

The approach was deliberately designed to be low-cost and reproducible — no specialist equipment, no telemetry, no laboratory analysis — so that it could be replicated at other coastal dune sites facing similar challenges.

Before the analysis, it helps to be clear about what the record actually contains. Figure 2 shows the full monthly observation history for the 66 wells of the core reference network: when each well came online, and for every month whether the reading was measured, interpolated across a single-month gap, recorded as dry or flooded, or simply missing. Monitoring began at a handful of wells in 2005 and was extended in stages, so the network only becomes near-complete in the early 2010s — one reason the change comparisons later in this summary rely on windows from that period onward. The two dashed vertical lines mark the April 2015 dune scraping and the December 2017 clearfell.

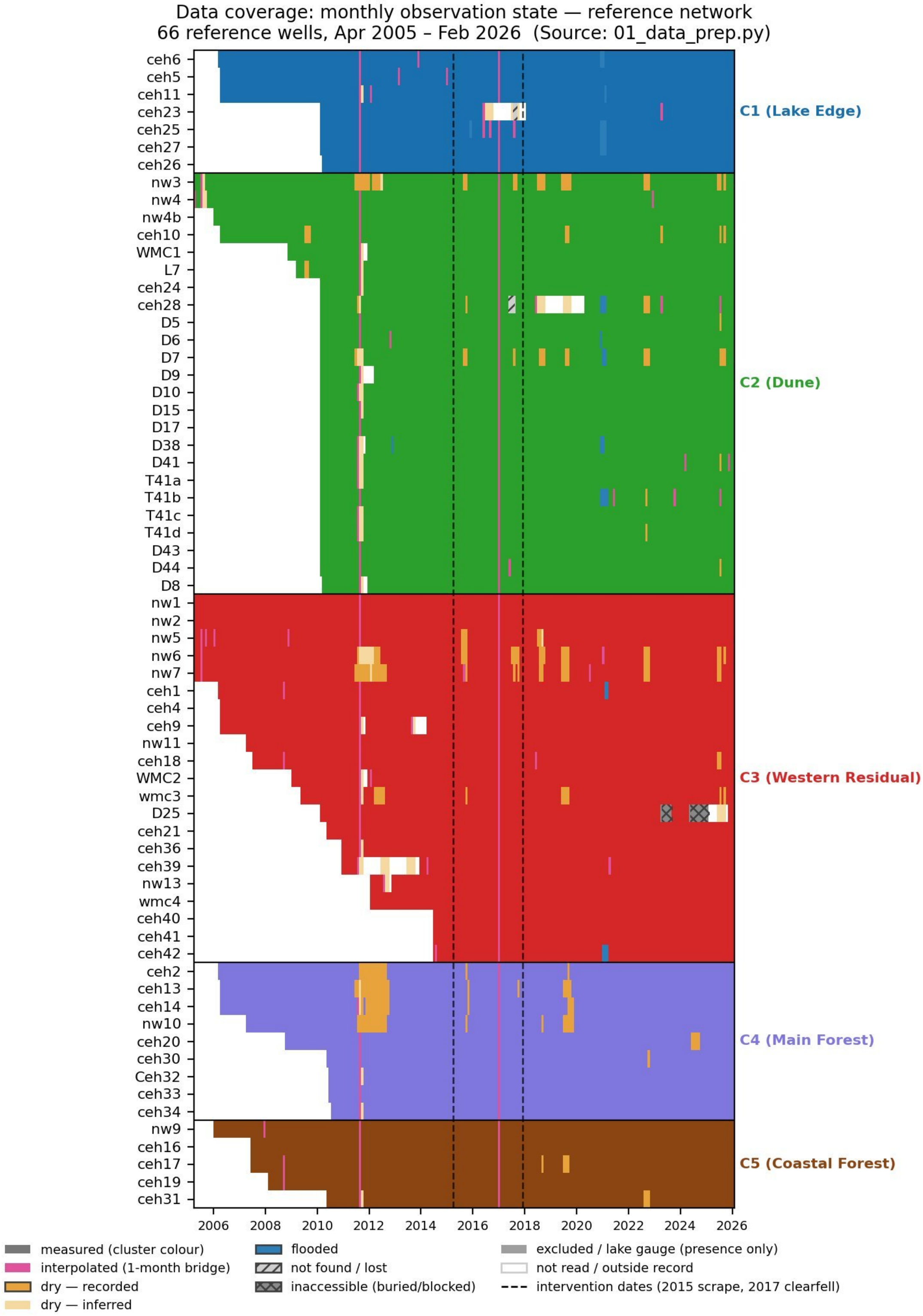

Figure 2. Data coverage for the reference network: the monthly observation state of all 66 reference wells, April 2005 to February 2026, grouped by zone. Colours distinguish measured, interpolated, dry, flooded and missing months; the dashed lines mark the 2015 scraping and 2017 clearfell. (Source: 01_coverage_states_reference.png; Figure 3a of the main report.)

How the model works

At the heart of the analysis is a simple water balance equation fitted independently at each well. Each month, the change in water level is explained by three competing forces: rainfall pushing the water table up, evaporation pulling it down, and drainage carrying water sideways towards the coast or lake. The model estimates the strength of each force for every well, producing three coefficients that together define how that part of the aquifer behaves. Wells with similar coefficients are grouped into zones.

This state-space model (SSM) was benchmarked against a simpler transfer function approach that lacks the drainage feedback term. Figure 3 shows both models against observed water levels at CEH6, a Dune cluster well. In one-step diagnostic mode (top panel) the two models are comparable, but in iterative forecasting mode (bottom panel) — where the model runs forward using only climate data and its own previous predictions — the SSM maintains realistic seasonal cycles (NSE = 0.66) while the transfer function diverges (NSE = −1.1). Across the network the SSM gave a positive forecasting score at 65 of 66 wells, against 44 of 66 for the simpler model. The drainage term is what prevents the model from drifting.

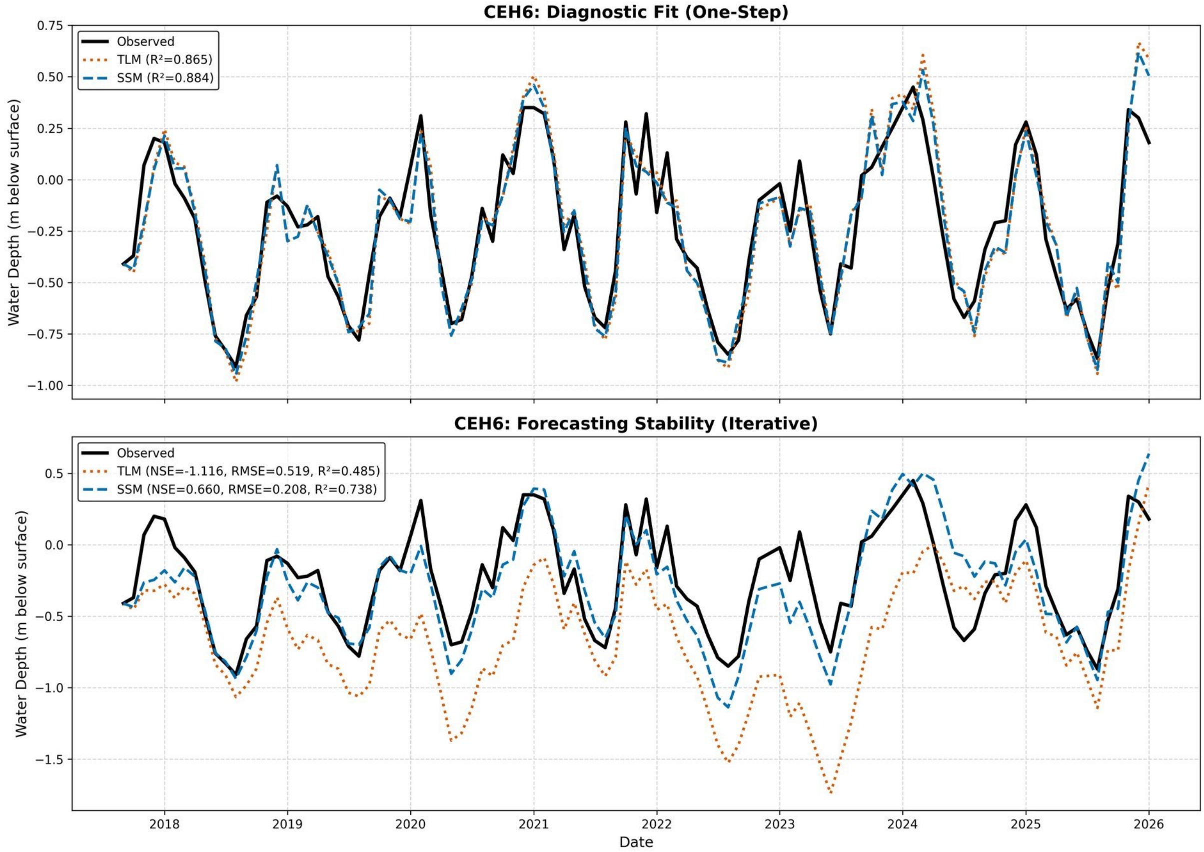

Figure 3. Model benchmarking at CEH6 (Dune cluster). Top: one-step diagnostic fit, where both the SSM (blue dashed) and transfer function (orange dotted) track the observed water level (black). Bottom: iterative forecasting, where the model runs forward using only climate forcing. The SSM maintains realistic amplitude and timing; the simpler model diverges progressively.

Seasonal prediction equations

Alongside the monthly SSM, a pair of seasonal prediction equations was fitted for each groundwater zone. A winter equation predicts the peak winter water level from cumulative winter rainfall (October–March) and the preceding summer minimum. A summer equation predicts the summer minimum from cumulative summer rainfall (April–September) and the preceding winter peak. Together these capture the aquifer's memory: how deep the summer drought was determines how much winter rain is needed to refill, and how high the winter peak reached determines how far the water table can fall the following summer.

For example, the Dune zone (C2) summer equation is:

$$\text{Summer minimum} = 0.0018 \times P(\text{summer}) + 0.54 \times h(\text{winter peak}) - 1.65 \quad (R^2 = 0.65)$$

This says: for every additional 100 mm of summer rain, the summer minimum is about 0.18 m shallower; and the higher the preceding winter peak, the shallower the following summer minimum. The intercept (−1.65 m) represents the structural deficit — even with average rainfall and an average winter peak, the summer minimum sits well below the ground surface.

Critical rainfall thresholds

Because the SSM is a physical mass balance rather than a statistical correlation, it can be run forwards: given a measured summer water level and a forecast of winter rainfall, the model calculates whether the water table will rise high enough to flood a given slack. This produces a critical rainfall threshold for each well — the amount of winter rain needed to achieve flooding, expressed as a multiple of the long-term average. Wells that need roughly twice the average winter rainfall or more are classified as structurally unreachable under any winter in the long climate record.

These equations are built into an interactive web tool (the scenario viewer) that allows site managers to explore how different management actions — tree felling, thinning, broadleaf conversion, dune scraping — and future climate scenarios would shift water levels across the entire site. A companion flood forecaster converts a single dipwell reading into a probability of winter flooding, updated as the season progresses.

Five distinct groundwater zones

The analysis identified five distinct zones across the site, each behaving differently depending on its geology, position and land cover:

- C1 — Lake Edge (eastern block): Shallow water table, responds quickly to rainfall, drains fast. The closest to the ecological tipping point.
- C2 — Dune (eastern block): Classic open dune, moderate rainfall response.
- C3 — Western Residual: Acts as a deep buffer — slower to respond, holds water longer.
- C4 — Main Forest: Pine plantation on thin soil over bedrock. Trees intercept rain and the shallow substrate amplifies summer drying.
- C5 — Coastal Forest: Pine plantation on deeper coastal sand. Shows the steepest water table decline of any zone — declining more than three times faster than the next worst.

A key finding is that it is the thickness of sand beneath each area — not the trees above — that controls how severely evaporation draws down the water table in summer. Within the forest zone, ground elevation alone explains about 95% of the variation in this drying effect.

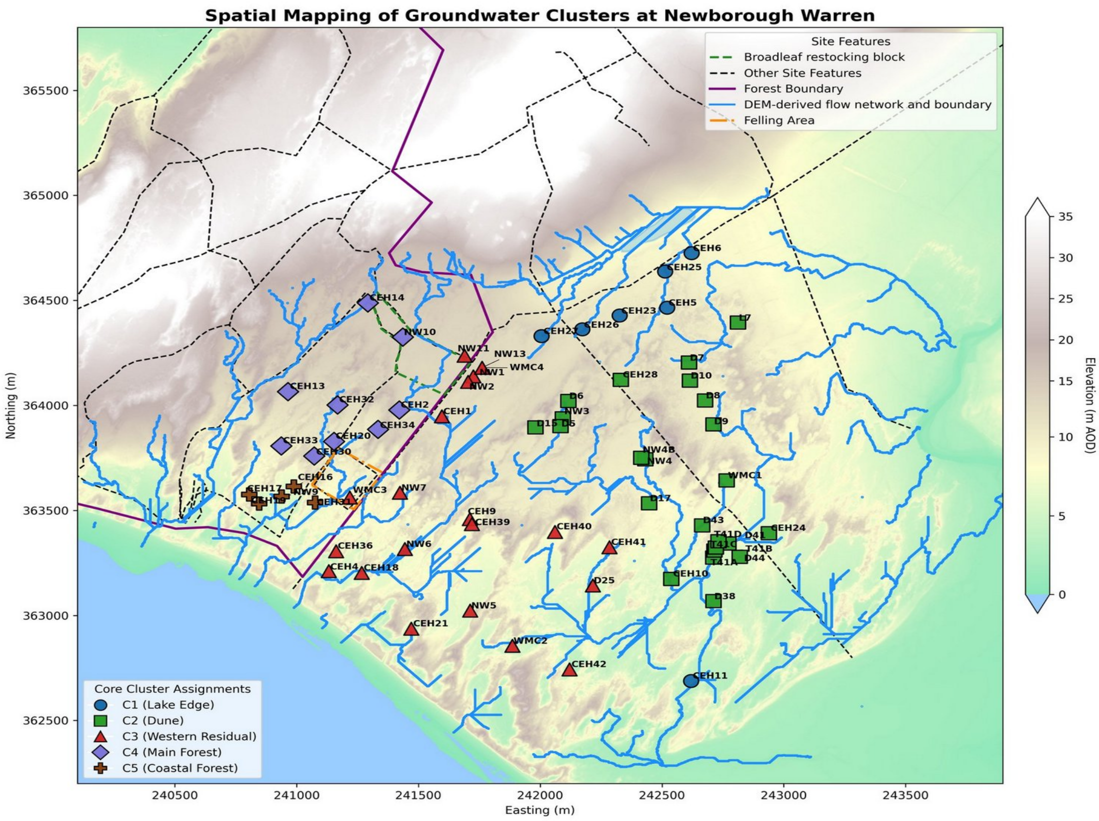

Figure 4. Spatial mapping of the five groundwater clusters identified by the analysis. Each colour represents a zone with distinct behaviour: C1 Lake Edge (blue), C2 Dune (green), C3 Western Residual (red), C4 Main Forest (purple), C5 Coastal Forest (brown).

How the five zones compare

The table below summarises the key characteristics of each groundwater zone. The ‘recharge sensitivity’ measures how strongly rainfall raises the water table; the ‘evaporation draw’ measures how strongly summer heat lowers it; the ‘drainage rate’ measures how quickly water drains away; and the ‘canopy share’ measures how much of the rainfall energy budget is consumed by evaporation from the canopy and soil rather than reaching the water table. The first three coefficients are dimensionless model parameters — larger values mean a stronger effect.

Zone	Character	Recharge sensitivity	Evaporation draw	Drainage rate (per month)	Canopy share
C1 Lake Edge	Shallow, fast-draining, near tipping point	4.58 (highest)	0.96 (lowest)	0.090 (fastest)	22%
C2 Dune	Open mature dune, moderate response	3.98	1.77	0.066	26%
C3 Western Residual	Deep buffer, slow response	3.57	1.85	0.058	28%
C4 Main Forest	Pine on thin substrate over bedrock	2.52 (lowest)	2.55 (highest)	0.020 (slowest)	40%
C5 Coastal Forest	Pine on deeper coastal sand; steepest decline	2.41	1.32	0.045	41%

Table 1. Summary of the five groundwater zones. Higher recharge sensitivity means a stronger water table response to rainfall; higher evaporation draw means more severe summer drying; higher canopy share means more rainfall intercepted by trees.

The central finding: summer is what matters

Conventional thinking assumes that winter rainfall is what fills dune slacks. This study found the opposite: it is the summer minimum water level that determines whether a slack will flood the following winter. If the water table drops too low during summer, no realistic amount of winter rain can refill the deficit.

This matters because summers are getting warmer. Since 2013, summer temperatures at the site have been nearly 1°C above the long-term average, increasing evaporation and deepening the summer drought.

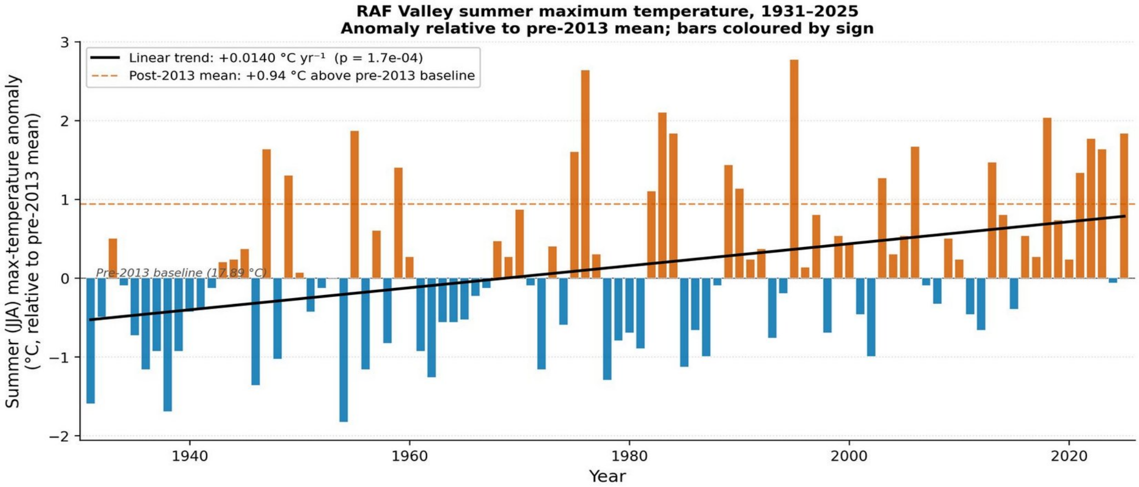

Figure 5. Summer temperature anomaly at RAF Valley (nearest weather station), 1931–2025. Orange bars show years warmer than the pre-2013 average; blue bars show cooler years. The dashed orange line marks the post-2013 mean: +0.94°C above baseline.

The water table has been declining across the network, with the trend statistically clearest at the Lake Edge (C1) and Coastal Forest (C5) zones; the Dune (C2) trend is marginal, and the Western Residual (C3) and Main Forest (C4) trends are not statistically significant on their own. Trend-line extrapolation indicates that the Lake Edge zone — where many of the most ecologically valuable slacks are located — is on course to cross the threshold for viable dune slack habitat around 2030–2032. Individual dry summers are already reaching that threshold.

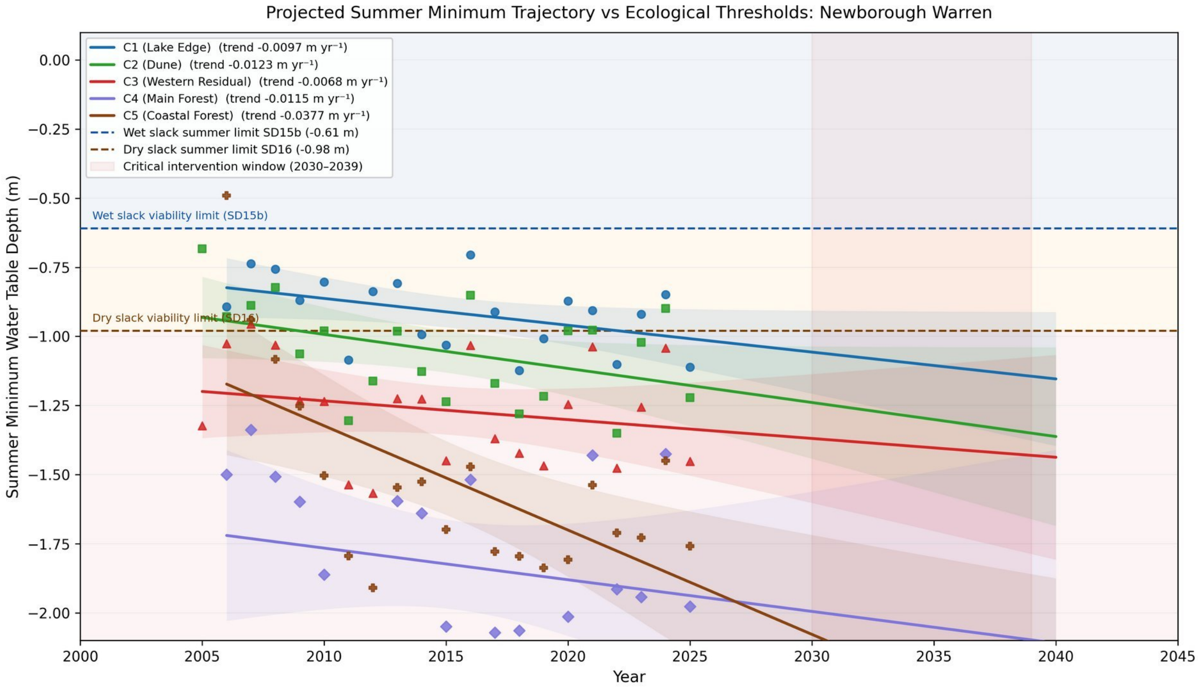

Figure 6. Projected summer minimum water table depth for all five clusters. The blue dashed line marks the wet slack threshold (−0.61 m); the brown dashed line marks the dry slack threshold (−0.98 m). The shaded band highlights the critical intervention window (2030–2039). All clusters are tracking downward.

A second measure: the spring baseline

The summer minimum is not the only way to read the health of a dune slack. A second measure — the five-year mean spring water level, or MSL5 — describes the sustained baseline within which slack vegetation actually establishes itself, averaged over the spring months and smoothed across five years to filter out single wet or dry seasons. Recent ecological work (van Willegen et al. 2025) identifies this spring baseline as the single best predictor of how the plant communities respond, which makes it a valuable companion to the summer minimum.

The two measures tell noticeably different stories. While the summer minimum is deepening sharply, the spring baseline is far more stable: it is projected to shift by only about 21–39 mm under the 2080s climate trajectory — three to five times less than the parallel deepening of the summer minimum. As a result the spring baseline stays within its present ecological regime at every zone, with the open-dune zones (C1, C2) sitting comfortably in the wet-slack range throughout the record and only the Coastal Forest (C5) sitting below the dry-slack threshold. Episodic threshold-crossings — where the water table falls below the dry-slack limit in dry summers — are therefore driven mainly by the summer minimum, while the spring baseline records the slower, sustained drift of the system as a whole.

The two measures track different aspects of the same ecological risk. The summer minimum determines whether the water table drops below the dry-slack threshold in any given year. The spring baseline is what can be measured most reliably across the whole network — summer readings often fail at deeper wells and winter readings are truncated by surface ponding — and in the published ecological work it correlates well with the vegetation community present at a slack. Both measures have been deteriorating, the summer minimum more steeply than the spring baseline.

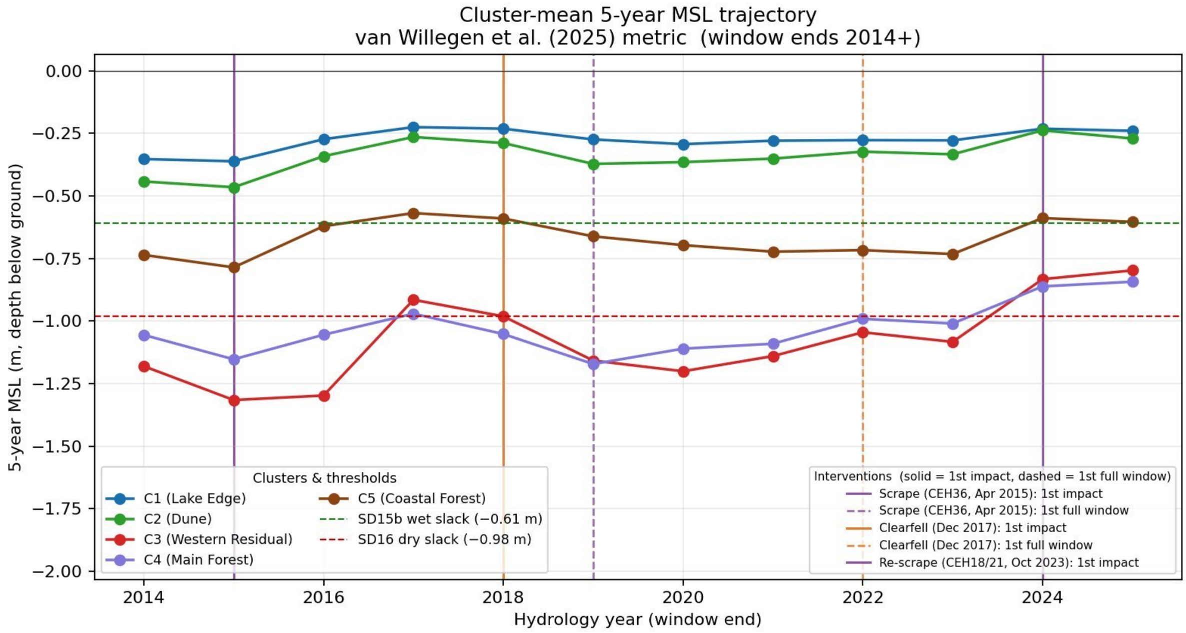

Figure 7. The five-year mean spring water level (MSL5) for each zone, 2014–2025. Compared with the summer minimum (Figure 6), the spring baseline is markedly flatter and more stable. The open-dune zones C1 (blue) and C2 (green) sit above the wet-slack threshold (green dashed); only the Coastal Forest (C5, brown) and parts of the western zones approach the dry-slack threshold (red dashed). Vertical lines mark the scraping and felling interventions.

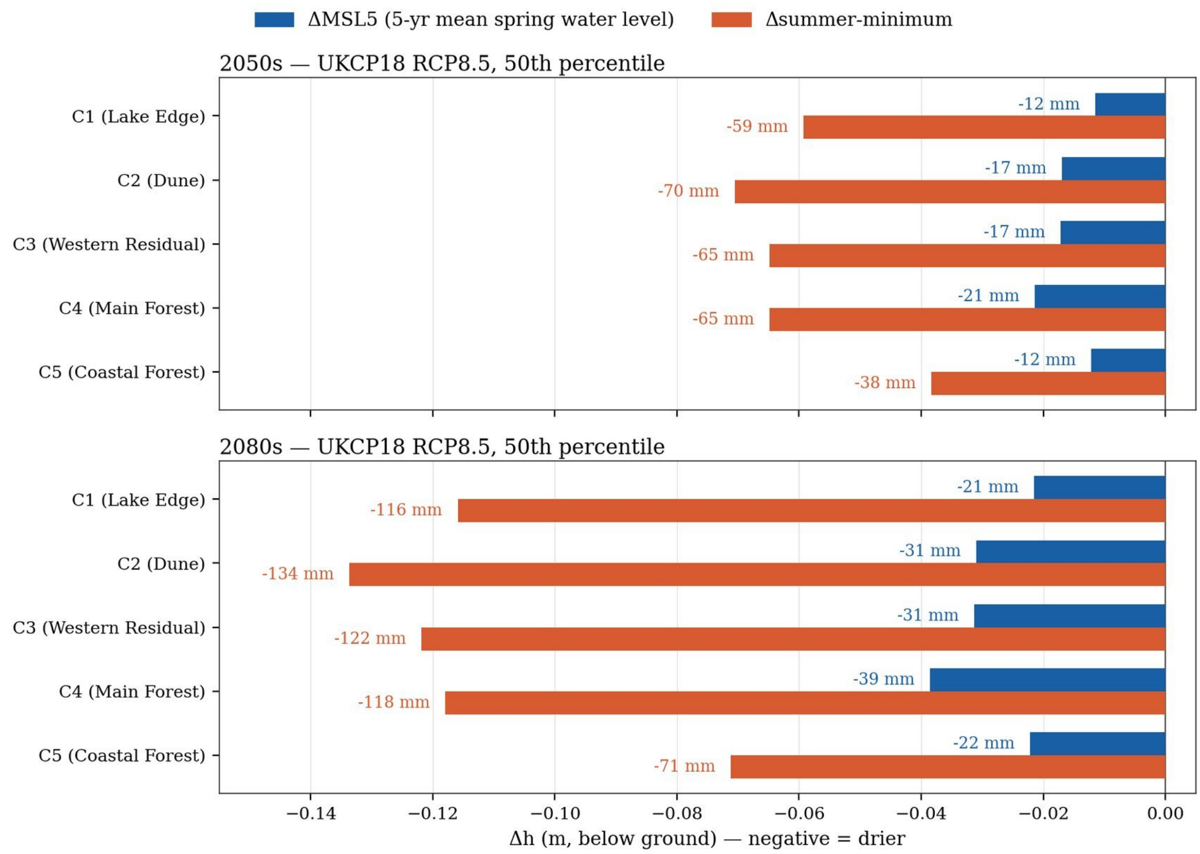

Figure 8. The projected drying of each zone by the 2050s and 2080s (UKCP18 high-emissions scenario), comparing the two ecological measures side by side. Blue bars show the shift in the spring baseline (MSL5); orange bars show the shift in the summer minimum. Bars extend left for a drier (deeper) water table. At every zone the summer minimum falls several times further than the spring baseline — by the 2080s the spring baseline drops by 21–39 mm while the summer minimum drops by 71–134 mm. It is the summer low point, not the year-round baseline, that bears the brunt of the projected climate change.

What management has been tried

Dune scraping

Dune scraping involves mechanically removing the top layer of sand to bring the ground surface closer to the water table. At one well (CEH36), three independent methods converged on a scraping benefit of roughly 80–140 mm, and the paired summer minimum shift against an unscraped control well was +195 mm (p = 0.004) — more than half of the critical 37 cm ecological gradient.

However, scraping only works where it is placed correctly in the landscape. Two further slacks, CEH18 and CEH21, were scraped much more recently, in October 2023, at more seaward sites. So far they show little measurable benefit — partly because barely two years of record is too short to judge an intervention against year-to-year weather, and partly because at these coastal locations the retreating shoreline is itself lowering the water table from the seaward edge inward, working against the scrape.

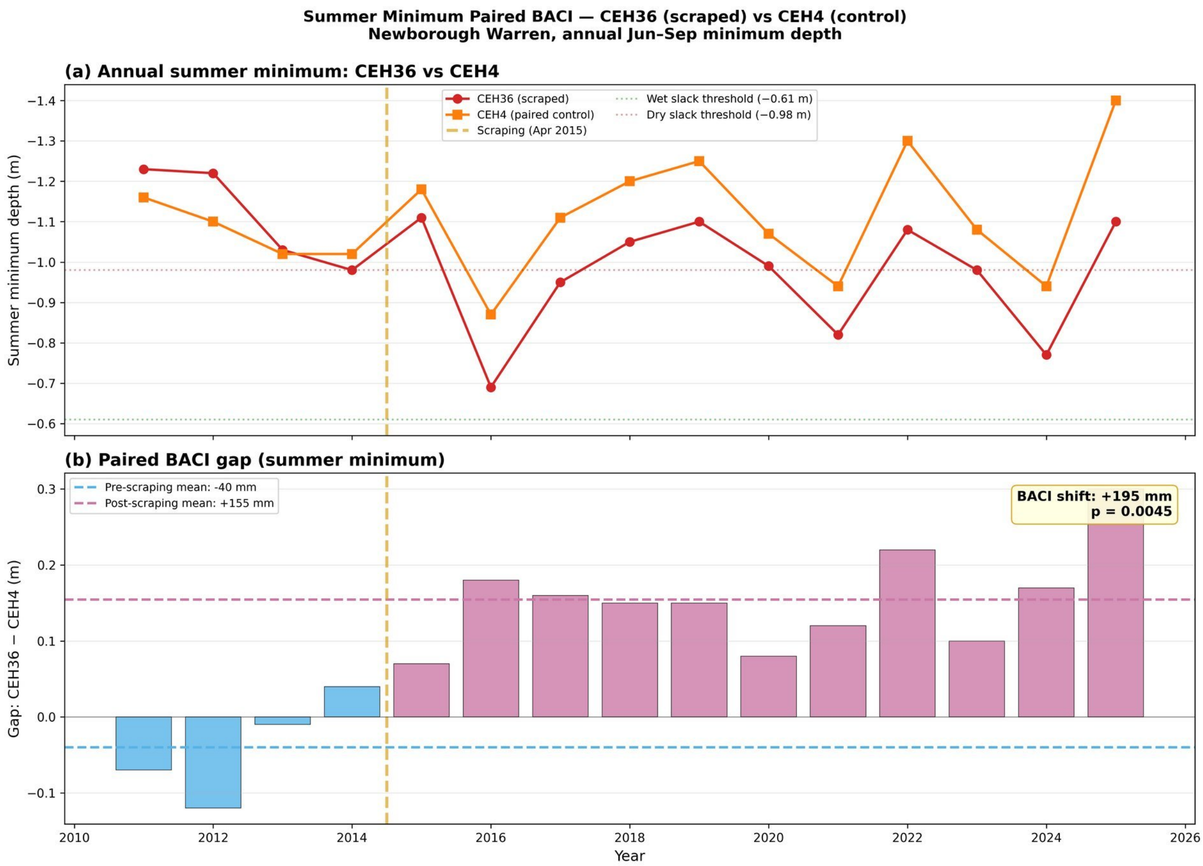

Figure 9. Summer minimum depth at the scraped well CEH36 (red) versus its paired control CEH4 (orange). Top: the two wells track together before scraping in 2015, then CEH36 sits consistently shallower. Bottom: the gap between them shifts from −40 mm before scraping to +155 mm after, a paired BACI shift of +195 mm (p = 0.004).

The reason scraping works at CEH36 is worth being precise about, because it sets a limit on what scraping can achieve elsewhere. The benefit is geometric: lowering the ground surface permanently moves it closer to the water table, so the same water table sits relatively higher beneath the slack. Scraping does not add water to the aquifer. This distinction matters, because the obvious alternative — trying to raise the water table itself by capturing more winter rainfall — runs into a hard physical constraint. The extra recharge steepens the local hydraulic gradient and drains away through the highly transmissive sand before the summer minimum is reached, so it provides little benefit at the time of year that determines whether a slack survives.

Scraping is therefore best understood as a targeted, local re-profiling of the ground rather than a way to recharge the aquifer. The same mechanism carries a less comfortable implication. Because a scrape works by intercepting water that would otherwise have drained further downgradient, the rise within the scraped pit is expected to be paid for, at least in part, by a small lowering of the water table just upslope of it — the scrape gaining by drawing water sideways from its inland neighbours rather than by adding water to the aquifer. If that is correct, it sets a limit on how far the technique can extend: each new scrape placed inland to recover the upslope loss would draw from its own inland neighbours in turn, so the front of lowered water table would creep inland with successive interventions. This remains an expected mechanism rather than a measured one — no drawdown has yet been detected at wells uphill of the scrapes, and the record since the 2023 interventions is still too short to confirm or rule it out. Either way, at well-chosen sites the local gain is gradually eroded by the background climate trend.

Tree felling

In December 2017, approximately 8.4 hectares of Corsican pine were clearfelled as a management experiment. The study used a rigorous five-tier experimental design with 17 wells and three independent control groups to test whether removing the trees raised the water table.

The result was nuanced. Comparing felled wells against unfelled forest — the most direct test — the analysis detected a statistically significant rise in mean monthly water levels of around +120 mm at the core impact well ($p < 0.001$). The rise of +33 mm at the forest edge was not statistically significant ($p = 0.19$), and a synthetic extension of within-compartment wells returned a step of +85 mm ($p < 0.001$). Crucially, the summer minimum depths — the metric that determines whether dune slacks survive — showed no corresponding improvement: felled and unfelled wells tracked each other through the summer. The clearfell raised average water levels but could not overcome the concurrent loss of recharge efficiency during the summer months when it matters most.

The analysis suggests the forest canopy acts as both a sink (intercepting rain) and a buffer (shielding the ground from direct evaporation in summer). Removing the canopy has more than one consequence. It recovers some winter recharge, because rain that the canopy used to intercept now reaches the ground. But it also raises the local water table relative to the surrounding forest, where canopy interception still suppresses recharge. The raised felled patch now sits higher than its forested neighbours, which steepens the local hydraulic gradient and drains the extra winter water sideways into the surrounding forest rather than holding it in place until summer. That faster drainage, combined with greater direct summer evaporation from the now-exposed soil, roughly cancels the winter recharge gain at the time of year that determines whether a slack survives — a point explored further below in the context of the site-wide decline in recharge efficiency.

Figure 10 makes this concrete. It shows the modelled seasonal water table cycle under the C4 baseline (blue) alongside three forest management options applied at the same site: full clearfell (red dashed), 50% thinning (orange) and broadleaf conversion (green dotted). The differences between the curves are small, and importantly, all four lines descend together into the dry-slack band during the summer window. Full clearfell raises the wet-season levels a little; thinning slightly less; broadleaf conversion almost not at all. None of them lifts the summer minimum back above the dry-slack threshold, and the summer trough is deeper than the wet-slack threshold in every case. The figure is the visual counterpart to the cancellation argument above: even with the canopy modelled away, the summer end of the cycle is set by other factors — direct evaporation, faster drainage of winter recharge, and the wider climate-driven decline — that no forest-management option addresses.

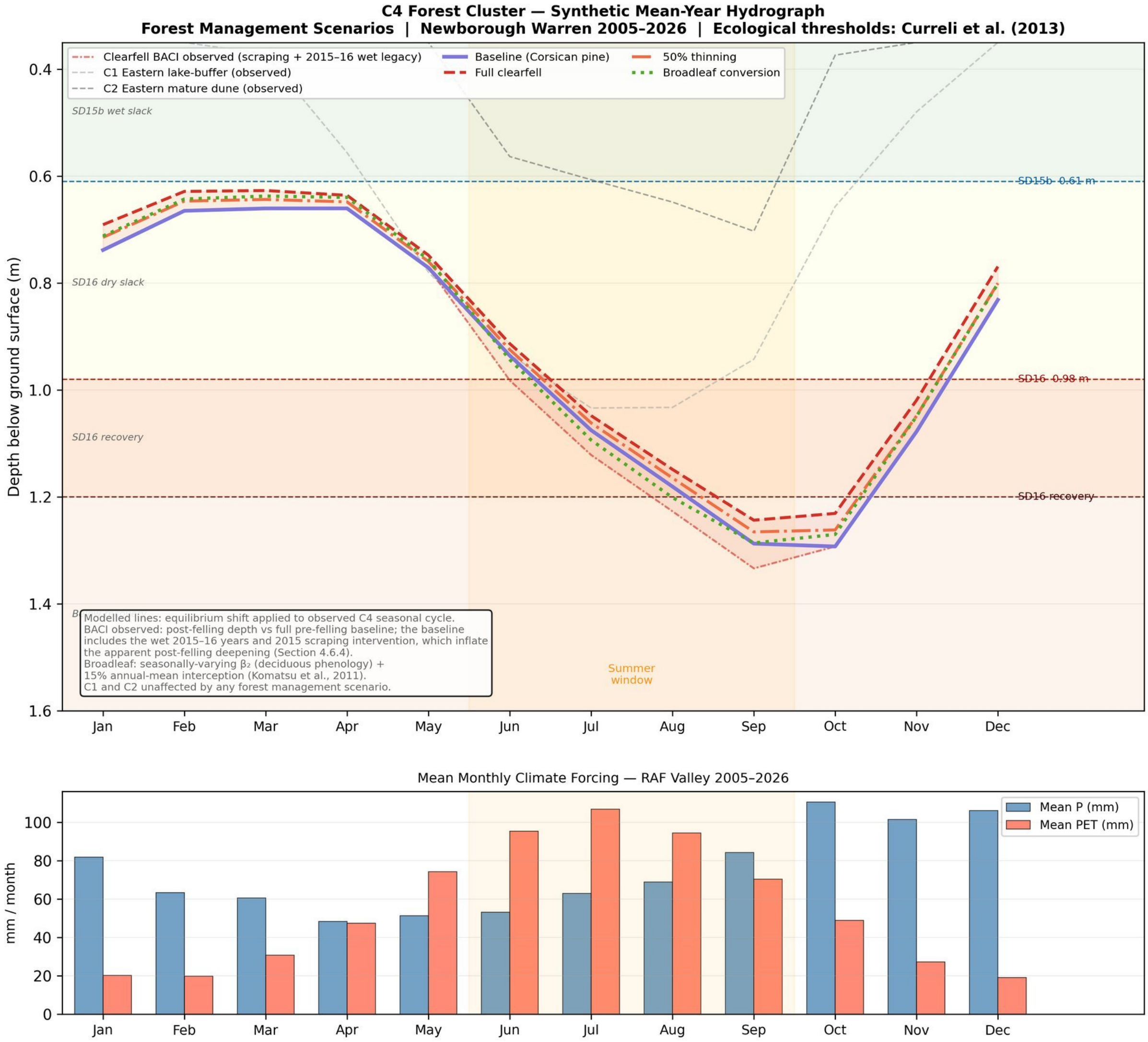

Figure 10. The seasonal water table cycle in the Main Forest zone (C4), showing the current baseline (blue) alongside modelled scenarios for full clearfell (red dashed), 50% thinning (orange) and broadleaf conversion (green dotted). The ecological threshold bands are shown as coloured horizontal zones. The bottom panel shows the seasonal pattern of rainfall versus evaporation.

Crucially, changes to the forest do not propagate sideways to the open dune slacks where the ecological need is greatest. The forest and the open dunes operate as largely independent groundwater systems, and the reason is the site's drainage geography: water beneath the forest flows south-westward towards Caernarfon Bay, while the open eastern dunes drain south-eastward towards the Menai Strait, with a topographic high separating the two. The forest's drawdown footprint, modelled separately, drops from over 50 mm within about 100 m of the forest edge to 1 mm or less at the eastern Lake Edge and Dune wells where the most valuable slacks lie. Forest management changes are carried away down the forest's own drainage path and never reach the eastern slacks.

Three drivers, three different footprints

It helps to see the three local drivers of water-table lowering on the same map and the same scale. Coastal erosion (illustrated as a single storm event) acts as a thin band along the shore but covers a large area. The forest canopy acts broadly and continuously, with a wide footprint concentrated around the plantation. A dune scrape acts intensely but only locally — large near the cut and falling away rapidly with distance. Crucially, none of them is a switch that can simply be flipped to reverse the others: the forest cannot push back coastal retreat, scraping cannot recover canopy interception, and the climate-driven background trend reaches across all three. The drivers overlap in some places and miss each other entirely in others, so the same intervention will land differently from one slack to the next.

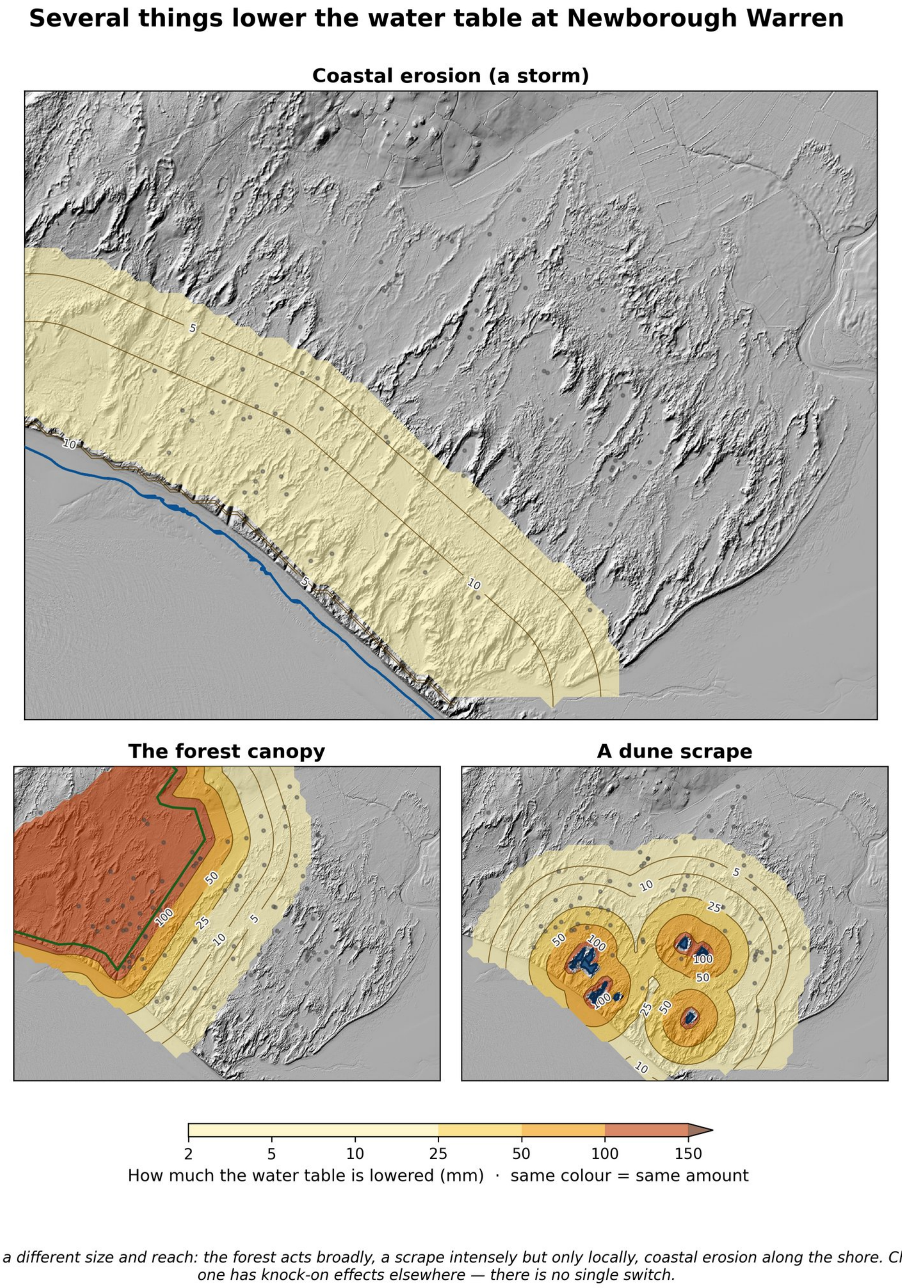

Figure 11. The three local drivers of water-table lowering shown together on a shared colour scale (same colour = same amount of drawdown, in millimetres). Top: coastal erosion from a single storm, spread along the shore. Bottom left: the forest canopy, acting broadly within and just outside the plantation. Bottom right: a dune scrape, intense but local. Each driver has a different size, shape and reach — there is no single intervention that addresses all three.

A declining recharge efficiency

Perhaps the most concerning finding is a site-wide decline in recharge efficiency — the proportion of rainfall that actually reaches the water table. This decline is the strongest candidate mechanism for the observed summer minimum deterioration across the entire monitoring network. It affects all five zones, regardless of whether they are forested or open, scraped or untouched, and operates independently of canopy cover. It is consistent with changing rainfall patterns (fewer sustained soaking events, more intense but brief storms that run off rather than infiltrating) and represents a climate-driven trend that no on-site management intervention can address.

This same trend explains why both interventions fall short in summer. When the trees were felled, the canopy stopped intercepting rain and winter recharge did increase — but that extra water steepened the local hydraulic gradient and drained away through the transmissive sand before the summer minimum arrived, just as it does after scraping. At the same time, removing the canopy exposed the soil to greater summer drying. The two effects roughly cancel out, so the felled wells track the unfelled forest through the summer. In the scenario modelling, clearfell and thinning produce only small positive net annual responses at the forest clusters (clearfell about +8 to +10 mm per month water-equivalent), while broadleaf conversion is close to neutral — all far smaller than the climate-driven changes. The common thread is that neither scraping nor felling can hold winter water in the aquifer until summer; the water table's summer low point is set by the site-wide decline in recharge efficiency, not by what is done to the surface or the canopy above it.

Coastal erosion: a hidden and lagged threat

The coastline at Newborough is retreating. As the beach erodes, the aquifer loses volume at its seaward edge, and the water table adjusts downward. A network-scale analysis attributes roughly 41% of the Coastal Forest zone's exceptional decline to this coastal-retreat signal, with the fitted effect reaching some hundreds of metres inland from the eroding western shoreline. Because groundwater moves slowly through sand, the effects of coastal erosion take years to propagate inland, so the decline visible in today's monitoring data partly reflects erosion that happened years ago. If erosion is accelerating, the worst effects have not yet arrived at interior wells.

Two control wells at different distances from the coast are both deteriorating in a pattern consistent with progressive coastal retreat, independently of any management action. Coastal-margin slacks may eventually require managed retreat rather than repeated restoration.

How water levels have actually changed

The spring water-level measurements span 2005 to the present, but the only pair of five-year windows clean enough to isolate the management interventions from climate noise are the ones ending in 2017 (springs 2013–2017, just before the clearfell) and 2023 (springs 2019–2023). Comparing them shows what has actually happened in the eight years that straddle the clearfell and the second-round scraping work. The result is a site-wide deepening of the spring baseline averaging about 97 mm across the network. Of the 59 wells with valid data in both windows, 56 deepened by more than 25 mm; none became shallower by more than 25 mm. The deepest losses are along the south-western coastal margin, exactly where the coastline is retreating; the eastern lake-edge area shows the smallest decline but still declined.

Two features of this picture deserve direct comment because they otherwise look inconsistent with numbers given earlier. First, the +120 mm clearfell rise reported in the management section is a comparison number — how much less the felled well dropped relative to unfelled forest wells on the same canopy and substrate — not a rise in absolute terms. The unfelled forest fell faster, the felled compartment fell more slowly, and +120 mm is the gap between them; both continue to track downward on the absolute scale shown here. Second, the starting point against which the felling effect was measured may itself have been depressed: the 2015 scrape, although located just outside the felled compartment, is expected to have drawn water sideways out of the surrounding ground, including the compartment itself, so the “before” water level was lower than it would have been without the scrape. Part of the recovery measured after felling could therefore be the felled compartment regaining ground it had already lost to the earlier scrape, rather than a fresh gain on top of an undisturbed baseline.

A later window ending in 2025 was examined but proved unusable: the exceptionally wet 2024 spring pulled the spring baseline sharply upward at slow-draining wells (especially the C4 Forest Controls, where water-table anomalies persist for years), masking the secular trend and making any apparent recovery a measurement artefact rather than a real ecological gain.

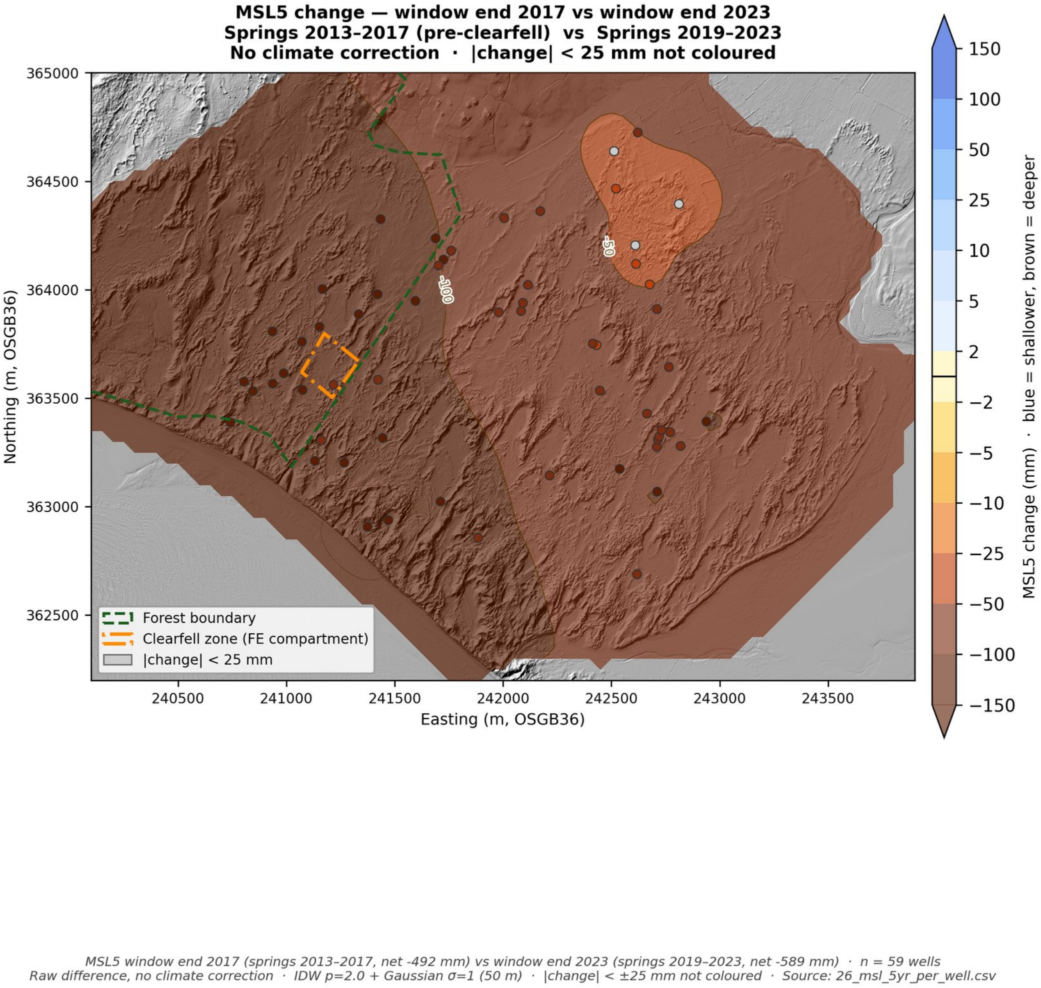

Figure 12. Observed change in the five-year spring water level baseline (MSL5) between window-end 2017 and window-end 2023, in mm (brown = deeper, blue = shallower). Wells with changes within ±25 mm of zero are left uncoloured. Across 59 wells in both windows, 56 deepened above the threshold and none became shallower; the largest declines are along the south-western coastal margin. The eastern lake-edge area (upper right) shows the smallest decline. The clearfell zone (orange) shows no signal distinguishable from the network-wide deepening. The 2015 scrape at CEH36 predates both comparison windows, so its initial rise does not appear here; what this period could in principle reveal is the slower inland propagation of the scrape’s drawdown, but any such signal would be small relative to — and inseparable from — the site-wide climate deepening. (Source: 20_msl5_change_2017_2023.png; reproduces Figure 58 of the main report.)

The site is not changing uniformly

The map in Figure 12 shows the overall picture: almost everything deepened. But looking at how the spring baseline is changing well by well, relative to the network as a whole, reveals something more nuanced — the site is being pulled in different directions at the same time.

Figure 13 shows this differential movement. Blue wells are holding up relative to the network average; red wells are sinking faster than average. Filled circles are individually statistically significant; hollow circles show the same direction but do not individually reach significance — the cluster-level pattern is consistent in both cases.

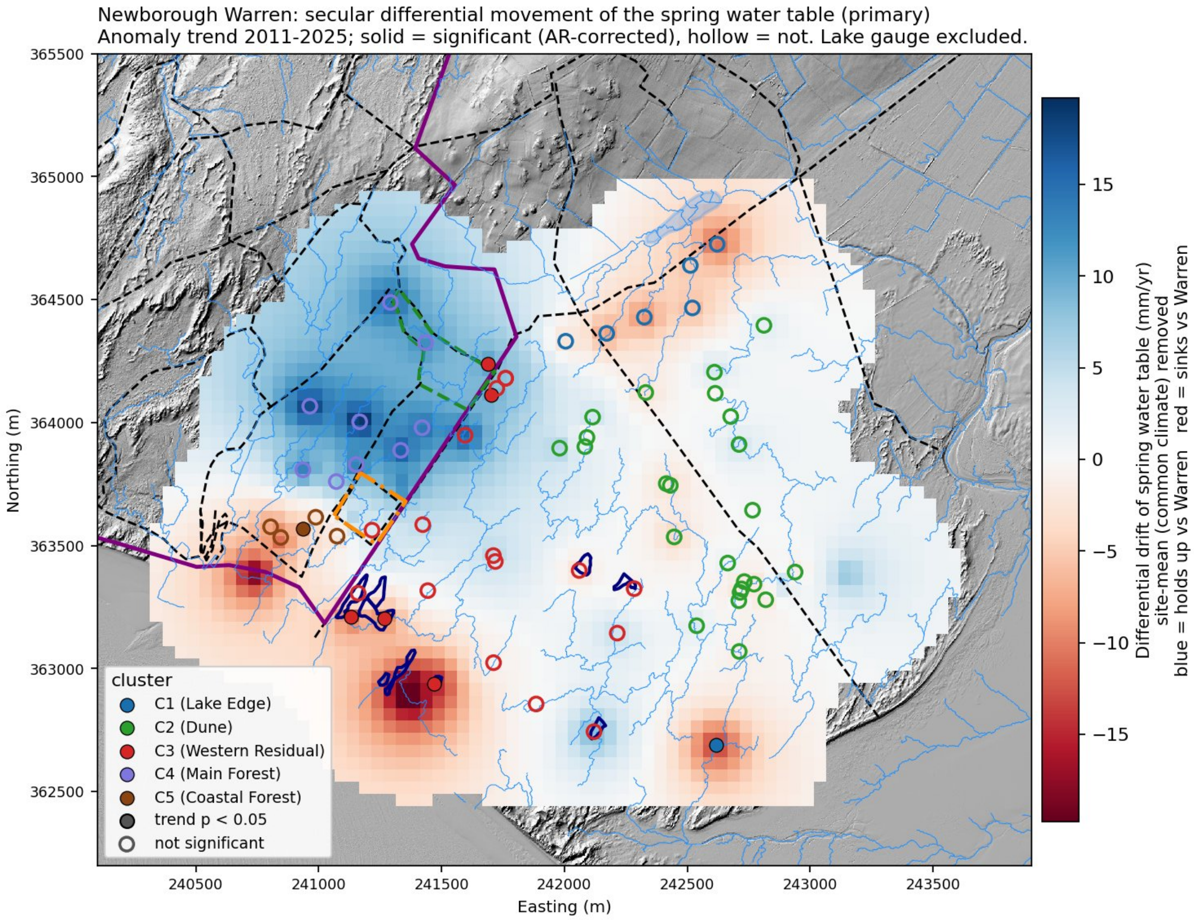

Figure 13. Differential movement of the spring water table, 2011–2025. Each well is shown as its trend relative to the site-mean spring level (mm per year); blue = rising relative to the network, red = sinking. Filled circles are individually significant (AR-corrected p < 0.05); hollow circles show the same directional tendency without reaching individual significance. The forest interior (C4, purple) is uniformly blue; the Lake Edge (C1, blue) and Coastal Forest (C5, brown) are uniformly red. The open dune zones (C2 green, C3 red) are broadly neutral. (Source: 32_differential_movement_2011_2025.png.)

The Main Forest zone (C4) is uniformly blue. Since 2011, spring water levels inside the Corsican pine plantation have been rising relative to the network average at around 15 mm per year across all nine reference wells. This looks, at first glance, like good news. It is not. Two things explain why this part of the aquifer swings more violently than the rest of the site. First, the forest occupies the topographic and hydraulic high of the aquifer — the area furthest from any lateral discharge boundary, whether the lake to the east, the Menai Strait to the south-east, or the coast to the south-west. Proximity to a constant-head boundary damps fluctuations: in a wet year the water table cannot rise far above the lake level near CEH6, and in a dry year it cannot fall far below it. The forest has no such backstop in any direction, so it rises freely in wet years and falls freely in dry ones. Second, the thin sand over bedrock beneath the forest gives it a low capacity to absorb recharge — the same property that makes summer droughts there so severe also concentrates any wet-year recharge into a larger head change. These two mechanisms reinforce each other, and together they produce the amplification visible in Figure 13. The forest is not recovering; it is oscillating more violently than the rest of the site, and a string of wet springs has temporarily pulled its average upward.

To put numbers on it: in the driest recent years (2011, 2012, 2019) the average spring depth in the Main Forest reaches around 1.7 m below ground. In the wettest years (2014, 2016, 2021, 2024) it rises to around 0.6 m — a swing of over a metre. The Lake Edge zone (C1) swings only about 0.5 m between the same extremes, buffered by the adjacent lake. The forest amplifies the climate signal at roughly 1.7 times the site average; the Lake Edge damps it to about 0.6 times.

While the forest interior rises in relative terms, the coastal zones are uniformly red. Every Lake Edge (C1) well and every Coastal Forest (C5) well is sinking relative to the network mean, at around 8 mm per year for the Lake Edge and 7 mm per year for the Coastal Forest. One well at the south-western coastal margin — outside the classified reference network — is declining at nearly 27 mm per year, the fastest in the dataset. These are the locations where the retreating shoreline is progressively removing aquifer volume. There is a further paradox here: despite carrying pine canopy like the Main Forest, the Coastal Forest wells swing far less between dry and wet years — at only about 0.9 times the site average. The coastal boundary prevents them from rising freely in wet years because the aquifer is shrinking at its seaward edge. They are caught: unable to benefit from wet winters, and continuing to fall as the shoreline retreats.

The open dune zones (C2 and C3) sit broadly neutral on the differential map, with individual wells moving in both directions. These are the zones closest to their ecological thresholds on the summer minimum measure, and they are also where targeted scraping remains the most tractable intervention. Their relative stability does not mean they are safe — it means the climate signal is reaching them uniformly, without the amplification that distinguishes the forest or the coastal loss that is draining the western margin.

The overall picture, then, is of a site being pulled apart: the forest interior temporarily buoyed by amplified wet-year responses, the coastal margin progressively drained by shoreline retreat, and the open dune zones — where the ecological need is greatest — drifting steadily deeper in absolute terms while holding roughly steady relative to each other. No single intervention addresses all three.

What this means for the future

The study's conclusions are sobering but clear:

1. The window for effective intervention is finite — measured in roughly one to two decades at the most vulnerable sites.

2. Targeted dune scraping at carefully chosen inland sites is the most effective direct intervention available, but its benefits are eroded over time by the background climate trend.

3. Tree felling raises average water tables in the forest but does not improve summer minima — the metric that determines dune slack survival. The evidence supports retaining forest cover while managing canopy density to partially recover the rainfall interception penalty without losing the summer buffering benefit. Forest management does not propagate to the open dune areas where habitat is most threatened.

4. Climate change is the binding constraint. Rising summer temperatures, declining recharge efficiency and coastal erosion are all operating in the same direction, and none can be addressed by on-site habitat management alone.

5. Continued monitoring is essential. The predictive tools developed in this study — including critical rainfall threshold equations and an interactive forecasting system — give site managers the ability to track whether conditions are on course to cross ecological thresholds, and to prioritise interventions where they can still make a difference.

Interactive web tools for Newborough

The analysis behind this summary has been packaged into three free web tools for Newborough. All three are accessible from a single project home page (newbroman.github.io/Newborough_Hydrology), which also hosts the full technical report this summary draws on, a more detailed methods supplement, and a user manual and technical note for the tools themselves.

The Hydrogeological Modelling Suite is an interactive scenario viewer that lets the user explore what happens to the water table at Newborough under different climate futures and management options, across all five zones. Pre-built presets cover UKCP18 climate projections, clearfell, thinning, and broadleaf conversion; results are shown as a map, a per-zone bar chart, and a downloadable table.

The Groundwater Flood Forecasting Tool is a per-well risk tool that takes the current state of the aquifer plus a forecast of winter rainfall and returns the probability that a given slack will flood this winter. The tool pulls live Met Office rainfall and updates as the season progresses, so a site manager can see how the prediction sharpens through autumn into the winter peak.

The Seasonal Extremes Scatter is a summary chart of every well in the monitoring network, plotting its mean annual summer minimum against its mean annual winter maximum. Wells are coloured by zone and the Curreli et al. (2013) ecological viability thresholds are overlaid, so the user can see at a glance which wells sit inside the viable wet-slack envelope and which sit outside it. A searchable index allows any individual well to be located within the scatter.

Together the three tools turn the analysis in this summary into a working set of decision-support instruments specific to Newborough. The same home page links onward to the underlying source code, the full monitoring dataset and the report itself for anyone wishing to look more deeply.